

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202512

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost_v1, Accuracy=0.9893713857357036, Precision=0.22254169062679702, Recall=0.911660777385159, F1=0.35775363993529, ROC_AUC=0.9860349941936518

algorithm=stacked_ensemble_v1, Accuracy=0.9547509408561026, Precision=0.06386020651310564, Recall=0.9469964664310954, F1=0.11965175980355681, ROC_AUC=0.9822966410407276

algorithm=easy_ensemble_v1, Accuracy=0.7619328091056513, Precision=0.010461018354622143, Recall=0.7726737338044759, F1=0.020642562698637464, ROC_AUC=0.852947825889866

algorithm=gradient_boosting_v1, Accuracy=0.9863537618945629, Precision=0.18169047061578084, Recall=0.9140164899882215, F1=0.303125, ROC_AUC=0.9835206145868565

algorithm=random_forest_v1, Accuracy=0.998978826913074, Precision=0.8523002421307506, Recall=0.8292108362779741, F1=0.8405970149253731, ROC_AUC=0.9839299458083961

algorithm=catboost_v2, Accuracy=0.99713153627268, Precision=0.5339739190116678, Recall=0.9163722025912838, F1=0.6747614917606245, ROC_AUC=0.9857038033190811

algorithm=stacked_ensemble_v2, Accuracy=0.9708487592938225, Precision=0.09622034100393466, Recall=0.950530035335689, F1=0.1747509744478129, ROC_AUC=0.9852749658227785

algorithm=easy_ensemble_v2, Accuracy=0.7652640516476455, Precision=0.01048170554171654, Recall=0.7632508833922261, F1=0.02067942110385984, ROC_AUC=0.8544423088930242

algorithm=gradient_boosting_v2, Accuracy=0.9885835143652664, Precision=0.21088251218191661, Recall=0.917550058892815, F1=0.3429451904028175, ROC_AUC=0.9845880859822671

algorithm=random_forest_v2, Accuracy=0.9988870360737998, Precision=0.8385922330097088, Recall=0.8138987043580683, F1=0.8260609683203826, ROC_AUC=0.9828619207710347